Accelerated Opinion Evolution for the Friedkin-Johnsen Model

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Abstract

Online social networks serve as a pivotal platform for opinion formation and dissemination, characterized by highly dynamic interaction patterns. The FriedkinJohnsen model provides a foundational framework for studying opinion dynamics in such settings. However, standard solvers for this model update opinions using only initial opinions and a fixed neighborhood structure, thereby ignoring temporal dependencies and suffering from slow convergence on large-scale graphs. To overcome this limitation, we propose a new class of efficient solvers that preserve the original equilibrium of the FriedkinJohnsen model while significantly accelerating convergence. Our approach enriches the update rule with temporal awareness by responding to the latest expressed opinions and retaining memory of past states. Experiments on real-world networks containing millions to tens of millions of nodes demonstrate the effectiveness, accuracy, and scalability of our methods.

Keywords

Opinion dynamics, social networks, graph algorithms, multi-agent systems

1 Introduction

Social media platforms have become an indispensable part of modern life, profoundly reshaping how individuals disseminate, exchange, and form opinions [? ? ?]. Compared to traditional channels of information diffusion, this transformation has significantly increased the complexity of opinion dynamics and exerted far-reaching effects on human behavior and social interaction patterns [?]. The pervasive nature of online interactions has spurred extensive academic interest in understanding the mechanisms underlying opinion propagation and formation, leading to the development of various mathematical models to capture these processes [? ? ? ?]. Among them, the FriedkinJohnsen (FJ) model [?] has emerged as one of the most widely applied frameworks. Notably, adaptations of the FJ model have been used to analyze complex negotiation processes, such as those leading to the Paris Agreement [?], demonstrating its utility in explaining consensus formation in multilateral international settings.

The core of the FJ model lies in its equilibrium opinion vector, which represents the stable set of expressed opinions reached after prolonged social influence. This equilibrium is determined jointly by the network topology, individuals intrinsic opinions, and interpersonal influence weights. It carries clear sociological meaning and serves as the foundation for quantifying key societal phenomena such as polarization [? ? ?], disagreement [? ?], conflict [?], and controversy [?]. However, existing solvers [? ? ?] typically implement opinion updates as a synchronous weighted average of initial opinions and neighbors opinions from the previous iteration. This approach implicitly assumes that individuals adjust their views based solely on outdated information from all neighbors, with no access to opinions updated earlier in the current round and no memory of their own past states. While this simplification facilitates theoretical analysis, it contradicts the temporal dependencies observed in real social interactions, where people often respond immediately to the latest statements and exhibit cognitive continuity over time. More critically, this static update scheme leads to extremely slow convergence on large-scale networks, severely limiting the practical applicability of the FJ model at real-world scales. Although numerous acceleration techniques exist for consensus-only models like DeGroot, they cannot be directly applied here. The FJ model explicitly preserves fixed intrinsic opinions, resulting in a more complex equilibrium structure that defies straightforward adaptation of existing methods.

To address these challenges, we propose a new class of efficient solvers that strictly preserve the original equilibrium of the FJ model while dramatically accelerating convergence through explicit modeling of temporal dynamics. Specifically, our approach captures responsiveness to the latest expressed opinions and retention of historical memory. It integrates three key mechanisms: sequential updates that immediately propagate freshly computed opinions within each iteration, self-interaction effects that enhance stability and enable faster convergence, and second-order historical dependence that incorporates past states to better capture the inertia of opinion evolution. We theoretically prove that all proposed algorithms converge to the same unique equilibrium as the classical model, yet with provably faster convergence rates. Experimental results demonstrate that our proposed algorithms exhibit superior performance and scalability on networks comprising tens of millions of nodes. The main contributions of this paper are summarized as follows:

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- We introduce a robust local iteration algorithm that efficiently approximates the equilibrium opinion vector while guaranteeing relative error bounds.
- By incorporating SOR techniques, we optimize the algorithm with a relaxation parameter ω, leading to faster convergence and improved computational efficiency.
- We conduct extensive numerical experiments on real-world network datasets. The results validate the advantages of our algorithms in terms of high efficiency and strong scalability.

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2 Related Work

The FJ model, integrating internal opinions with social influence, stands as a pivotal framework in social dynamics research [**?**]. Stability conditions for this model were established in [**?**], while equilibrium expressed opinion was derived in [**? ?**]. Interpretations of the FJ model were further explored in [**?**] and [**?**]. Research has also delved into adversarial interventions [**? ? ?**] and the spread of viral content [**?**]. Furthermore, the FJ model has been expanded in various dimensions, as discussed in [**? ?**], including multidimensional approaches [**? ?**].

Beyond the foundational aspects, interpretations, and extensions of the FJ model, researchers have developed metrics to quantify disagreement [**?**], polarization [**? ?**], conflict [**?**], and controversy [**?**]. These metrics offer valuable insights into social dynamics. Various optimization problems based on the FJ model have been addressed, including efforts to minimize overall opinion [**?**], disagreement [**?**], polarization [**? ?**], and conflict [**? ?**] by adjusting node attributes such as internal opinions or susceptibility to persuasion [**? ? ?**]. Network modifications, like edge additions or deletions to reduce polarization or foster consensus, have emerged as a significant area of study [**? ? ?**].

Scalability and efficiency pose significant challenges in large networks, where conventional methods like matrix inversion are impractical. To overcome this, scalable algorithms utilizing sampling and distributed computing have been introduced [**? ?**], facilitating the analysis of networks with millions of nodes. For undirected graphs, Laplacian solvers can approximate equilibrium opinions in near-linear time [**?**], while directed graphs rely on forest sampling methods [**?**]. Recent work [**?**] propose sublinear time algorithms to estimate single-node opinions without computing the entire equilibrium vector.

3 Preliminaries

In this section, we introduce some useful notations and fundamental concepts of the FJ model of opinion formation for the convenience of description of problem formulation and algorithm.

3.1 Notations

We use $\mathbb{R}$ to denote the set of real numbers, and regular lowercase letters (e.g., a, b, c) to represent scalars in $\mathbb{R}$. Bold lowercase letters (e.g., $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$) denote column vectors, and bold capital letters (e.g., $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$) denote matrices. For a matrix $\boldsymbol{A}$, A_{ij} denotes its (i, j)-th entry, and for a vector $\boldsymbol{a}$, a_i denotes its i-th element. The transpose of a vector or matrix is denoted by $\boldsymbol{a}^\top$ and $\boldsymbol{A}^\top$, respectively. We use $\mathbf{1}$ to denote the all-ones vector of appropriate dimensions, and $\boldsymbol{e}_u$ to denote the u-th standard basis vector. The identity matrix of appropriate dimensions is denoted by $\boldsymbol{I}$. For a square matrix $\boldsymbol{A}$, we respectively denote its strictly lower-triangular and strictly upper-triangular parts by $\boldsymbol{A}_{\text{low}}$ and $\boldsymbol{A}_{\text{up}}$.

3.2 Graph and Matrix Definitions

Let $\mathcal{G} = (V, E)$ be an unweighted simple graph with $n = |V|$ nodes and $m = |E|$ edges, where V is the set of nodes and $E \subseteq V \times V$ is the set of edges. The graph $\mathcal{G}$ is either a directed graph or an undirected graph depending on the context. The adjacency relation of all nodes in $\mathcal{G}$ is encoded in the adjacency matrix $\boldsymbol{A}$, whose entry $a_{ij} = 1$ if i and j are adjacent, and $a_{ij} = 0$ otherwise. For any given node i, N_i denote the set of its neighbors, meaning $N_i = \{j : (i, j) \in E\}$. The degree d_i of any node i is defined by $d_i = \sum_{j=1}^{n} a_{ij}$. The degree diagonal matrix of $\mathcal{G}$ is $\boldsymbol{D} = \mathrm{diag}(d_1, d_2, \cdots, d_n)$. The degree vector $\boldsymbol{d}$ is given by $\boldsymbol{d} = \boldsymbol{D}\mathbf{1}$. The transition matrix of an unbiased random walk on graph $\mathcal{G}$ is $\boldsymbol{P} = \boldsymbol{D}^{-1}\boldsymbol{A}$, which is a row-stochastic matrix. The Laplacian matrix $\boldsymbol{L}$ of $\mathcal{G}$ is $\boldsymbol{L} = \boldsymbol{D} - \boldsymbol{A}$, which is a symmetric semi-positive definite matrix. Matrix $\boldsymbol{L}$ is singular, and thus it is irreversible. For a matrix $\boldsymbol{M}$, if the magnitudes of all the eigenvalues of M are less than 1, then the series $\sum_{k=0}^{\infty} \boldsymbol{M}^k$ is a Neumann series and converges to $(\boldsymbol{I} - \boldsymbol{M})^{-1}$.

3.3 Friedkin–Johnsen (FJ) Model

As one of the first opinion dynamics models, the FJ model is an extension of the DeGroot's opinion model. In the DeGroot model, every node has only one opinion that is updated as the weighted average of its neighbors. Different from the DeGroot model, in the FJ model, each node $i \in V$ has two different kinds of opinions: one is the internal (or innate) opinion s_i, the other is the expressed opinion z_i. The internal opinion s_i is assumed to remain constant, private to node i, while the expressed opinion z_i evolves as a weighted average of its corresponding internal opinion s_i and the expressed opinions of i's neighbors.

Given a graph $G = (V, E)$, let $\mathbf{s}$ denote the internal opinion vector, with each entry satisfying $s_i \in [0, 1]$. The formulation of the equilibrium opinion expressed in the FJ model is an iterative form such that

$$z_i^{(t+1)} = \frac{s_i + \sum_{j \in N_i} a_{ij} z_j^{(t)}}{1 + \sum_{j \in Ni} a_{ij}}. \tag{1}$$

As in the FJ iterative formula above, the opinion of each node in the FJ model is updated as the weighted average of the neighbor's expressed opinion z_j and its own internal opinion s_i. When t approches infinity, the updated expressed opinion was shown to converge to a unique equilibrium opinion $\boldsymbol{z}_i$. So the iterative update in Equation 1 can be written compactly as $\boldsymbol{z}^{(t+1)} = (\boldsymbol{I} + \boldsymbol{D})^{-1}\left(\boldsymbol{s} + \boldsymbol{A}\boldsymbol{z}^{(t)}\right)$ in vector form, which expresses the FJ dynamics as a linear transformation applied at each iteration. Then the system of equilibrium expressed opinion vector $\boldsymbol{z}$ is equivalent to: $(\boldsymbol{I} + \boldsymbol{L})\boldsymbol{z} = \boldsymbol{s}$, where $\boldsymbol{I}$ is the identify matrix , and $\boldsymbol{L} = \boldsymbol{D} - \boldsymbol{A}$ is the Laplacian matrix. When $\mathcal{G}$ is connected and $a_{ij} \geq 0$, the matrix $\boldsymbol{I} + \boldsymbol{L}$ is symmetric positive definite, ensuring existence and uniqueness of $\boldsymbol{z}$. This characterization motivates estimators and algorithms that exploit the spectral and structural properties of the Laplacian.

3.4 Game-theoretic Interpretation

From a game-theoretic perspective, the classical FJ model admits an equivalent interpretation as the equilibrium of a quadratic utility game. Recall that the FJ equilibrium satisfies $(\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A})\boldsymbol{z}=\boldsymbol{s}$. For undirected graphs, the matrix $\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}$ corresponds to a symmetric, positive definite Laplacian-based operator, which allows the equilibrium to be characterized as the unique minimizer of a strongly convex quadratic objective.

This formulation further admits a decentralized interpretation. Updating z_i according to the FJ rule is equivalent to each agent i minimizing the local quadratic cost

$$(z_i-s_i)^2+\sum_{j\in N(i)} w_{i,j}(z_i-z_j)^2, \tag{2}$$

given the opinions of its neighbors. Thus, opinion averaging can be viewed as a sequence of best-response updates in a complete-information game, where each agent selects an opinion to balance conformity with its intrinsic belief. The resulting Nash equilibrium coincides with the minimizer of the global social cost and is unique due to strong convexity.

3.5 Exsiting Algorithm

In this subsection, we introduce existing algorithms for estimating the equilibrium opinion vector. Since direct matrix inversion is computationally infeasible due to its (3) complexity, alternative methods are necessary. Algorithms for directed graphs are discussed in Appendix B.

Algorithm Based on Laplacian Solver. The authors in [?] proposed an approximation algorithm based on approximate Laplacian solver [?]. By leveraging the positive semidefiniteness of $(\boldsymbol{I}+\boldsymbol{L})$, the algorithm avoids direct matrix inversion and instead approximates the solution with rigorous error guarantees. For the equilibrium opinion vector, given an error parameter ϵ, the algorithm computes an approximate solution $\hat{\boldsymbol{z}}$ in $O\left(m\log^3 n\log\left(\frac{\sqrt{n}}{\epsilon}\right)\right)$ time, satisfying

$$(1-\epsilon)\|\boldsymbol{z}\|_2^2\le\|\hat{\boldsymbol{z}}\|_2^2\le(1+\epsilon)\|\boldsymbol{z}\|_2^2.$$

This algorithm demonstrates strong performance on large-scale undirected graphs, particularly for million-level networks. However, when the number of network nodes is relatively small, its performance is suboptimal due to the fixed overhead of preprocessing steps, such as graph sparsification and recursive decomposition. These steps are necessary for achieving efficiency on large graphs but become disproportionately costly for relatively small graphs. Additionally, due to its space complexity requirements, the algorithm cannot scale to graphs with more than tens of millions of edges. And it is important to note that this algorithm is specifically designed for undirected graphs and does not extend to directed graphs.

4 Accelerated Opinion Iteration

We study accelerated iterative schemes for computing the equilibrium of the Friedkin–Johnsen (FJ) model. Beyond the classical synchronous update, we consider sequential and latest-opinion-based iterations that immediately exploit newly updated opinions. These schemes are further extended with self-interaction, yielding a family of accelerated algorithms that preserve the same equilibrium as the standard FJ model while achieving faster convergence. Their convergence properties are analyzed in the following subsections.

4.1 Sequential Latest-Opinion Updates

The classical FJ iteration updates all opinions synchronously, using only the opinion profile from the previous iteration. In contrast, we consider a sequential update scheme in which opinions are revised one by one, and newly updated opinions are immediately used in subsequent updates.

Under this scheme, opinions are updated in a coordinate-wise manner. Specifically, when updating the i-th opinion at iteration $t+1$, the most recent values of opinions $1,\dots,i-1$ are used whenever available, while the remaining opinions retain their values from iteration t:

$$z_i^{(t+1)}=\frac{s_i+\sum_{j=1}^{i-1}a_{ij}z_j^{(t+1)}+\sum_{j=i+1}^{n}a_{ij}z_j^{(t)}}{1+\sum_{j\in N_i}a_{ij}}. \tag{3}$$

This update reduces to the classical synchronous FJ iteration when all opinions on the right-hand side are taken from iteration t. As shown later, the sequential latest-opinion update preserves the FJ equilibrium and typically converges faster due to the immediate utilization of newly updated opinions. The sequential latest-opinion update admits a compact matrix representation. Specifically, it can be written as $\boldsymbol{z}^{(k+1)}=\boldsymbol{B}\boldsymbol{z}^{(k)}+\boldsymbol{c}$, where $\boldsymbol{B}=(\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})^{-1}\boldsymbol{A}_{up}$, $\quad \boldsymbol{c}=(\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})^{-1}\boldsymbol{s}$.

Algorithm 1 implements the proposed sequential latest-opinion iteration. At each iteration, the algorithm performs a single forward sweep over the nodes in a fixed order. When updating node i, the most recent opinion values are used whenever available, i.e., newly updated values from the current sweep are immediately incorporated, while the remaining opinions are taken from the previous iterate. This latest-available policy enables faster information propagation along the sweep and distinguishes the method from the classical synchronous FJ iteration.

The update combines the aggregated neighbor influence with the internal opinion s_i and applies a local normalization by $1+d_i$. The iteration terminates when a prescribed tolerance is met or a maximum number of iterations is reached. Let $\epsilon^{(k)}=\boldsymbol{z}^{(k)}-\boldsymbol{z}$ denote the error at iteration k, where $\boldsymbol{z}$ is the FJ equilibrium. We next establish the convergence of the sequential latest-opinion iteration. The theorem below establishes the convergence of the latest-value updates.

Theorem 4.1 (Convergence and Linear Rate of Latest-Opinion Iteration). *Consider the latest-opinion iteration*

$$\boldsymbol{z}^{(t+1)}=(\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{\text{low}})^{-1}\boldsymbol{A}_{\text{up}}\boldsymbol{z}^{(t)}+(\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{\text{low}})^{-1}\boldsymbol{s}, \tag{4}$$

The iteration converges linearly to the unique FJ equilibrium opinion $\boldsymbol{z}=(\boldsymbol{I}+\boldsymbol{L})^{-1}\boldsymbol{s}$ for any initial opinion $\boldsymbol{s}$.

Proof. Let $\boldsymbol{z}^* = (\boldsymbol{I}+\boldsymbol{L})^{-1}\boldsymbol{s}$ denote the true value estimated by FJ model, and define the error $\epsilon^{(k)} := \boldsymbol{z}^{(k)} - \boldsymbol{z}^*$. Then the error propagation satisfies $\epsilon^{(k+1)} = \boldsymbol{B}\epsilon^{(k)}$, Since matrix $\boldsymbol{I}+\boldsymbol{L}$ is strictly diagonally dominant by rows, i.e., $|1+d_i| > \sum_{j\neq i}|a_{ij}|, \quad 1 \le i \le n$, it is an L-matrix and hence nonsingular. Moreover, the matrix $\boldsymbol{M} := \boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low}$ is nonsingular.

We estimate the error in the ℓ_1-norm. For each component i, the Latest-opinion update implies

$$|\epsilon_i^{(k+1)}| \le \frac{1}{|1+d_i|}\left(\sum_{j=1}^{i-1}|a_{ij}|\,|\epsilon_j^{(k+1)}| + \sum_{j=i+1}^{n}|a_{ij}|\,|\epsilon_j^{(k)}|\right). \tag{5}$$

Using the definition of the ℓ_∞-norm, we obtain $\|\epsilon^{(k+1)}\|_1 \le \frac{\sum_{j=i+1}^{n}|a_{ij}|}{|1+d_i|-\sum_{j=1}^{i-1}|a_{ij}|}\|\epsilon^{(k)}\|_1$. By the strict diagonal dominance condition, the denominator is strictly larger than the numerator. Define $q := \max_i \frac{\sum_{j=i+1}^{n}|a_{ij}|}{|1+d_i|-\sum_{j=1}^{i-1}|a_{ij}|}$, then $0 \le q < 1$ and

$$\|\epsilon^{(k+1)}\|_1 \le q\,\|\epsilon^{(k)}\|_1. \tag{6}$$

Iterating (6) yields $\|\epsilon^{(k)}\|_1 \le q^k\,\|\epsilon^{(0)}\|_1$, which implies $\epsilon^{(k)} \to 0$ as $k \to \infty$. Therefore, the latest-opinion iteration converges linearly to $\boldsymbol{z}^*$. □

The parameter q not only guarantees convergence but also characterizes the speed of convergence. Smaller values of q imply faster stabilization of opinions, which is particularly important for large-scale networks. The next theorem is going to be useful in showing the main result.

Theorem 4.2 (Improvement over Classical FJ Iteration). *The latest-value iteration converges faster than the classical FJ iteration.*

Proof. The FJ iteration matrix is $\mathbf{B}_{\mathrm{FJ}} = (\mathbf{I}+\mathbf{D})^{-1}\mathbf{A}$, , while the latest-value iteration matrix is $\mathbf{B}_{\mathrm{LI}} = (\mathbf{I}+\mathbf{D}-\mathbf{A}_{\mathrm{low}})^{-1}\mathbf{A}_{\mathrm{up}}$. By the Stein–Rosenberg theorem for nonnegative matrix splittings (see, e.g., [Thm. 3.33]varga2000matrix), the spectral radii of the corresponding iteration matrices satisfy $\rho(\mathbf{B}_{\mathrm{LI}}) \le \rho(\mathbf{B}_{\mathrm{FJ}}^2) = \rho(\mathbf{B}_{\mathrm{FJ}})^2$. Since $0 < \rho(\boldsymbol{B}_{\mathrm{FJ}}) < 1$, we have $\rho(\boldsymbol{B}_{\mathrm{FJ}})^2 < \rho(\boldsymbol{B}_{\mathrm{FJ}})$, which implies $\rho(\boldsymbol{B}_{\mathrm{LI}}) < \rho(\boldsymbol{B}_{\mathrm{FJ}})$.

It is well known that for stationary linear iterations, the asymptotic convergence rate is characterized by the spectral radius of the iteration matrix (see, e.g.,[Ch. 2]varga2000matrix). □

Lemma 4.3 (Error Bound). *Let $\boldsymbol{z}^{(t+1)}$ denotes the estimate updated by the latest-opinion iteration at epoch $t+1$, and $\boldsymbol{r}^{(t)} = \boldsymbol{s} - (\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A})\boldsymbol{z}^{(t)}$ denotes the residual error. Then, for all $t \ge 0$, the error bound between the estimate opinion and the true opinion satisfies*

$$\|\boldsymbol{z}^* - \boldsymbol{z}^{(t)}\|_1 \le (n+m)\|\boldsymbol{z}^{(t+1)} - \boldsymbol{z}^{(t)}\|_1, \tag{7}$$

Proof. By the LatestIter update,

$$(\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})\boldsymbol{z}^{(t+1)} = \boldsymbol{A}_{up}\boldsymbol{z}^{(t)} + \boldsymbol{s}.$$

Algorithm 1: LatestIter($G, \boldsymbol{s}, T_{max}, \epsilon$)

Input: A graph $G=(V,E)$, an internal opinion $\boldsymbol{s}$; maximum iteration limit $T_{\max}$; tolerance ϵ;
Output: Estimated opinion vector z; iteration count t

```
z ← 0 ∈ ℝ^n
for t ← 1 to T_max do
    z^old ← z
    for i ← 1 to n do
        foreach j ∈ N_i do
            if j < i then
                h ← h + z_j
            else
                h ← h + z_j^old
        z_i ← (s_i + h)/(1 + d_i)
    if ‖z − z^old‖_1 < ε/(n+m) then
        return (z, t)
return (z, it)
```

Subtracting $(\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})\boldsymbol{z}^{(t)}$ from both sides yields

$$(\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})(\boldsymbol{z}^{(t+1)} - \boldsymbol{z}^{(t)}) = \boldsymbol{s} - (\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A})\boldsymbol{z}^{(t)} = \boldsymbol{r}^{(t)}.$$

Hence,

$$\boldsymbol{z}^{(t+1)} - \boldsymbol{z}^{(t)} = (\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})^{-1}\boldsymbol{r}^{(t)}.$$

On the other hand, the fixed point satisfies $\boldsymbol{z}^* = (\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A})^{-1}\boldsymbol{s}$, which implies

$$\boldsymbol{z}^* - \boldsymbol{z}^{(t)} = (\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A})^{-1}\boldsymbol{r}^{(t)}.$$

Combining the two relations gives

$$\boldsymbol{z}^* - \boldsymbol{z}^{(t)} = (\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A})^{-1}(\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})(\boldsymbol{z}^{(t+1)} - \boldsymbol{z}^{(t)}).$$

Let $\boldsymbol{z}^{t+1} - \boldsymbol{z}^{t} \le \epsilon\mathbf{1}$, take the ℓ_1 norm and using submultiplicativity,

$$\|\boldsymbol{z}^* - \boldsymbol{z}^{(t)}\|_1 \le \epsilon\|(\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A})^{-1}\|_1\|(\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})\mathbf{1}\|_1.$$

For the standard Friedkin–Johnsen setting, $(\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A})$ is strictly diagonally dominant, implying $\|(\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A})^{-1}\|_1$ by the Varah bound. Moreover,

$$\|(\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})\mathbf{1}\|_1 = \sum_{j=1}^{n}(1+d_j - \sum_{i=j+1}^{n}a_{ij}) \le n+m$$

which completes the proof. □

The iterative scheme yields geometrically decaying residuals, and the estimation error converges to zero. In the following, we will provide a time complexity analysis

Theorem 4.4 (Time Complexity of LatestIter). *To achieve an error $\|z^{(T)} - z\|_1 \le \varepsilon$, the time complexity is bounded by:*

$$O\left\lceil m\frac{\ln(\frac{\varepsilon}{\|z_1 - z_0\|_1}(1-q))}{\ln(q)}\right\rceil.$$

PROOF. By standard linear convergence theory for stationary iterations (e.g., [? ?]), the residuals decay geometrically. In particular, for any vector norm $\|\cdot\|$ induced by a matrix norm $\|\cdot\|_v$, let $q = \|\boldsymbol{B}\|_v < 1$. Then for all $t \geq 1$,

$$\|\boldsymbol{z}-\boldsymbol{z}^{(t)}\|_1 \;\leq\; \frac{q}{1-q}\,\|\boldsymbol{z}^{(t)}-\boldsymbol{z}^{(t-1)}\|_1 \;\leq\; \frac{q^t}{1-q}\,\|\boldsymbol{z}^{(1)}-\boldsymbol{z}^{(0)}\|_1, \tag{8}$$

where the first inequality follows from the Neumann series expansion of $(\boldsymbol{I}-\boldsymbol{B})^{-1}$, and the second inequality follows by induction.

To achieve an absolute error $\|\boldsymbol{z}^{(T)} - \boldsymbol{z}\|_1 \leq \varepsilon$, it suffices to choose T such that $\frac{q^T}{1-q} \leq \frac{\varepsilon}{\|z_1-z_0\|_1}$, which yields

$$T \;=\; \left\lceil \frac{\ln(\frac{\varepsilon}{\|z_1-z_0\|_1}(1-q))}{\ln(q)} \right\rceil .$$

Each iteration of LatestIter involves a single sparse matrix–vector multiplication with $\boldsymbol{B}$, which costs $O(m)$ time, where m is the number of edges in the graph. Therefore, the overall time complexity to reach the desired accuracy is

$$O(m \cdot \lceil c_1 \ln(\varepsilon + c2\rceil),$$

where $c_1 = \frac{1}{ln(q)}, c_2 = \frac{ln\frac{(1-q)}{\|z_1-z_0\|_1}}{\ln(q)}$, completing the proof. □

4.2 Self-Interaction Opinion Updates

The sequential latest-opinion update enables rapid information propagation; however, it relies on the strong assumption that agents completely overwrite their opinions at each update, which limits its ability to capture persistent self-influence. In many realistic scenarios, agents retain a fraction of their own opinions and adjust them incrementally rather than fully replacing them. To account for such self-interaction effects, we augment the latest-opinion update and propose a self-interaction mechanism that blends the previous opinion with the latest sequential estimate. Specifically, let $\tilde{\boldsymbol{z}}^{(t+1)}$ denote the intermediate opinion produced by the LatestIter scheme.

The self-interaction update is defined as $\boldsymbol{z}^{(t+1)} = (1-\beta)\boldsymbol{z}^{(t)} + \beta\tilde{\boldsymbol{z}}^{(t+1)}$, where β is a control parameter that governs the strength and direction of self-interaction. Under this update, each agent combines its own opinion with the latest sequential estimate produced by the LatestIter scheme. Equivalently, the component-wise update for agent i can be written as

$$z_i^{(t+1)} = (1-\beta)z_i^{(t)} + \beta\frac{1}{1+d_i}\left(s_i + \sum_{j=1}^{i-1} a_{ij}z_j^{(t+1)} + \sum_{j=i+1}^{n} a_{ij}z_j^{(t)}\right).$$

This formulation preserves the equilibrium of the original Friedkin–Johnsen model, while explicitly incorporating self-interaction into the iterative process. As a result, it improves numerical stability and, in practice, often leads to faster convergence.

Based on the self-interaction update rule, we propose an accelerated iterative method, termed SelfIter. The algorithm introduces a single control parameter to regulate self-interaction, while preserving the locality of updates and the linear structure of the original iteration. As a result, SelfIter remains computationally lightweight and compatible with large-scale networks, while providing improved numerical behavior over the vanilla latest-opinion update. Algorithm 2 presents SelfIter as a refinement of the latest-order scheme in Algorithm 1, designed to enhance robustness and convergence under challenging conditions. By construction, SelfIter reduces to the LatestIter iteration when $\beta = 1$.

Algorithm 2: SelfIter($G, \boldsymbol{s}, \beta, T_{max}, \epsilon$)

```
Input: A graph G = (V, E), an internal opinion s;
       self-interaction parameter β; maximum
       iteration limit T_max; tolerance ϵ
Output: Estimated opinion vector z; iteration count t
z ← 0 ∈ ℝ^n;   z^old ← z;
for t ← 1 to T_max do
    z^old ← z
    for i ← 1 to n do
        h ← 0
        foreach j ∈ N_i do
            if j < i then
                h ← h + a_ij z_j
            else
                h ← h + a_ij z_j^old
        y_i ← (s_i + h) / (1 + d_i)
        z_i ← z_i + β (y_i − z_i)
    if ‖z − z^old‖_∞ ≤ ϵ then
        return (z, t)
return (z, t)
```

The self-interaction update admits the following matrix form: $\boldsymbol{z}^{(k+1)} = \big((\boldsymbol{I}+\boldsymbol{D}) - \beta\boldsymbol{A}_{\text{low}}\big)^{-1}\big((1-\beta)(\boldsymbol{I}+\boldsymbol{D}) + \beta\boldsymbol{A}_{\text{up}}\big)\boldsymbol{z}^{(k)} + \beta\big((\boldsymbol{I}+\boldsymbol{D}) - \beta\boldsymbol{A}_{\text{low}}\big)^{-1}\mathbf{s}$, where $\boldsymbol{A}_{\text{low}}$ and $\boldsymbol{A}_{\text{up}}$ denote the strictly lower and upper triangular parts of $\boldsymbol{A}$, respectively. This representation makes explicit that SelfIter induces an affine linear iteration. We next analyze its convergence properties and derive sufficient conditions on β to guarantee convergence.

LEMMA 4.5 (CONVERGENCE RANGE FOR SELFITER). *$\boldsymbol{I} + \boldsymbol{D} - \boldsymbol{A}$ has nonzero diagonal entries so that the update in SelfIter is well defined. Then the self-interaction iteration converges for all $0 < \beta < 2$.*

PROOF. We first derive a necessary condition. If the SelfIter iteration converges, its iteration matrix $\boldsymbol{G}$ must satisfy $\rho(\boldsymbol{G}) < 1$, where $\rho(\cdot)$ denotes the spectral radius. A basic property of eigenvalues implies $|\det(\boldsymbol{G})| = \prod_{i=1}^n |\lambda_i| \leq \rho(\boldsymbol{G})^n < 1$, and hence $|\det(\boldsymbol{G})|^{1/n} < 1$.

Using the explicit matrix form of SelfIter, we obtain

$$\det(\boldsymbol{G}) = \det(\boldsymbol{I}+\boldsymbol{D}-\beta\boldsymbol{A}_{\text{low}})^{-1}\det\big((1-\beta)(\boldsymbol{I}+\boldsymbol{D})+\beta\boldsymbol{A}_{\text{up}}\big).$$

Since both matrices are triangular with identical diagonal entries, their determinants can be computed directly, yielding $\det(\boldsymbol{G}) = (1-\beta)^n$. Therefore, $|1-\beta| < 1$, which is equivalent to $0 < \beta < 2$.

We now establish sufficiency. As $\boldsymbol{I} + \boldsymbol{D} - \boldsymbol{A}$ is strictly diagonally dominant, the original Friedkin–Johnsen system admits a unique equilibrium, and the corresponding linear operator defines a contraction under an appropriate vector norm. The SelfIter update combines the current own iterate with the latest-order update through a convex interpolation parameter β. For $0 < \beta < 2$, this interpolation preserves the contraction property of the underlying opinion update operator, which implies $\rho(\boldsymbol{G}) < 1$. Consequently, the SelfIter iteration converges.

□

The parameter β admits a clear interpretation in terms of opinion adjustment. When $\beta \in (0,1)$, SelfIter performs a conservative correction, moderating the update toward the latest locally computed estimate. The choice $\beta = 1$ recovers the sequential latest-opinion update, while $\beta \in (1,2)$ induces an over-corrective behavior that amplifies the update step and can further accelerate convergence.

Accordingly, β can be viewed as an *adjustment gain* that regulates the magnitude of correction applied at each iteration. The update can be equivalently written as

$$\boldsymbol{z}^{(t+1)} = \boldsymbol{z}^{(t)} + \beta\big(\tilde{\boldsymbol{z}}^{(t+1)} - \boldsymbol{z}^{(t)}\big),$$

which highlights that SelfIter applies a gain-controlled adjustment toward the latest local estimate while preserving the locality of updates.Moreover, we derive the optimal choice of β that minimizes the convergence rate of the memory-augmented algorithm.

Lemma 4.6 (Optimal Gain for Fastest SelfIter Convergence). *Let $\boldsymbol{B} = \boldsymbol{I} + \boldsymbol{D} - \boldsymbol{A}$ denote the iteration matrix of the latest-opinion update, with spectral radius $\rho(\boldsymbol{B}) = \rho \in (0,1)$. Then, for the SelfIter iteration matrix $\mathcal{L}_\beta$, the spectral radius $\rho(\boldsymbol{G})$ is minimized over $\beta \in (0,2)$ at*

$$\beta^* \triangleq \frac{2}{1+\sqrt{1-\rho^2}} \in (1,2).$$

Proof. Since $\boldsymbol{I} + \boldsymbol{D} - \boldsymbol{A}$ is symmetric positive definite, the latest-opinion iteration is a convergent linear fixed-point scheme. Its iteration matrix B_0 is diagonalizable with real eigenvalues contained in $(-1,1)$, and thus $\rho(B_0) = \rho \in (0,1)$.

The SelfIter update applies a gain-controlled correction to the latest-opinion estimate. Consequently, the error dynamics can be analyzed mode-wise: for each eigenvector of B_0 associated with eigenvalue $\mu \in (-\rho, \rho)$, the corresponding error component evolves independently under a linear recurrence induced by $\mathcal{L}_\beta$.

A direct calculation shows that the induced update on each spectral mode admits two characteristic multipliers $\lambda_\pm(\beta,\mu)$, which are the roots of

$$\lambda^2 - (1-\beta)\lambda - \frac{\beta^2}{4}\mu^2 = 0. \tag{9}$$

Solving (9) yields

$$\lambda_\pm(\beta,\mu) = \frac{1}{2}\Big(1-\beta \pm \sqrt{(1-\beta)^2 + \beta^2\mu^2}\Big).$$

Therefore, the spectral radius of $\mathcal{L}_\beta$ is given by

$$\rho(\mathcal{L}_\beta) = \max_{\mu\in[-\rho,\rho]} \max\big\{|\lambda_+(\beta,\mu)|,\ |\lambda_-(\beta,\mu)|\big\}.$$

For any fixed $\beta \in (0,2)$, the above expression is maximized at $|\mu| = \rho$, corresponding to the slowest-decaying error mode. Hence,

$$\rho(\mathcal{L}_\beta) = \frac{1}{2}\left|1-\beta + \sqrt{(1-\beta)^2 + \beta^2\rho^2}\right|.$$

To minimize $\rho(\mathcal{L}_\beta)$ over $\beta \in (0,2)$, the optimal choice equalizes the magnitudes of the two characteristic roots, which yields

$$\beta^* = \frac{2}{1+\sqrt{1-\rho^2}}.$$

Substituting β^* into the above expression gives $\rho(\mathcal{L}_{\beta^*}) = \frac{1-\sqrt{1-\rho^2}}{1+\sqrt{1-\rho^2}}$. Since $\rho \in (0,1)$, it follows that $\beta^* \in (1,2)$ and $\rho(\mathcal{L}_{\beta^*}) \in (0,1)$. This completes the proof. □

5 Opinion Iteration with Historical Memory

In the previous section, we developed accelerated opinion update schemes by refining the iterative perspective of the classical Friedkin–Johnsen (FJ) model. In particular, the *LatestIter* framework exploits the most recently available neighbor information, while the gain-controlled variant further modulates the relative impact of newly propagated opinions and agents' current states. These designs are motivated by practical considerations in real-world opinion diffusion and remain within the standard first-order dynamical setting, where opinion evolution depends solely on the current configuration.

Beyond the iterative interpretation, the FJ model admits an alternative and equally important explanation from a game-theoretic perspective. Specifically, it can be viewed as a quadratic utility game in which each agent balances social conformity against attachment to its intrinsic opinion, and the resulting equilibrium corresponds to a Nash equilibrium that minimizes a global social cost. This viewpoint highlights the FJ dynamics as an implicit equilibrium-seeking process driven by rational, yet bounded, decision making.

However, existing studies predominantly focus on a Markovian opinion updating mechanism, where the state transition depends only on the current opinion profile. Such memoryless dynamics neglect the influence of historical opinions, despite the fact that, in realistic social interactions, individuals often revise their beliefs by accounting for recent experience and accumulated past stances. From a game-theoretic standpoint, this limitation excludes an important class of path-dependent behaviors that naturally arise in repeated or evolving games, especially under noisy environments or gradual belief stabilization.

Motivated by this observation, we extend the FJ opinion dynamics by incorporating a finite memory into the update process. Rather than relying exclusively on the current opinion state, the proposed model allows agents to leverage their immediate opinion history, leading to a second-order opinion iteration. This modification preserves the linear structure of the original FJ model, while enriching its temporal dynamics and enabling a more faithful representation of history-dependent strategic adjustment. As will be shown, the resulting scheme admits a compact state-space formulation, which facilitates convergence analysis and spectral characterization.

While this interpretation justifies the convergence of standard FJ dynamics, it also reveals an inherent limitation: the induced best-response process is memoryless and depends only on the current opinion profile. In contrast, many repeated and evolving games exhibit path-dependent behavior, where agents incorporate past actions to stabilize or accelerate convergence. This observation motivates the incorporation of historical opinions into the FJ dynamics, leading naturally to a second-order, memory-augmented opinion evolution scheme.

Recall that the classical FJ iteration updates the opinion vector according to $\boldsymbol{z}^{(t+1)} = (\boldsymbol{I}+\boldsymbol{D})^{-1}\boldsymbol{A}\boldsymbol{z}^{(t)} + (\boldsymbol{I}+\boldsymbol{D})^{-1}\boldsymbol{s}$, where $\boldsymbol{z}^{(t)} = [z_1^{(t)}, \dots, z_n^{(t)}]^\top$ denotes the opinion profile at iteration t.

In the memory-augmented setting, given the current and previous opinion states $\boldsymbol{z}^{(t)}$ and $\boldsymbol{z}^{(t-1)}$, we first compute the intermediate updates produced by the FJ operator, $\boldsymbol{x}^{(t)} = (\boldsymbol{I}+\boldsymbol{D})^{-1}\boldsymbol{A}\boldsymbol{z}^{(t)} + (\boldsymbol{I}+\boldsymbol{D})^{-1}\boldsymbol{s},\ \boldsymbol{x}^{(t-1)} = (\boldsymbol{I}+\boldsymbol{D})^{-1}\boldsymbol{A}\boldsymbol{z}^{(t-1)} + (\boldsymbol{I}+\boldsymbol{D})^{-1}\boldsymbol{s}$. The next opinion state is then obtained by forming a convex combination of these two intermediate states,

$$\boldsymbol{z}^{(t+1)} = \eta\, \boldsymbol{x}^{(t)} + (1-\eta)\, \boldsymbol{x}^{(t-1)},$$

where $\eta \in (0,1)$ controls the relative influence of the most recent and historical information. The initial condition is set as $\boldsymbol{z}^{(0)} = \boldsymbol{z}^{(-1)}$.

From an implementation perspective, this scheme requires each agent to retain only the locally aggregated opinion from the previous iteration, rather than the entire opinion trajectory. As a result, the memory overhead remains minimal while enabling richer temporal dynamics. When $\eta = 1$, the iteration reduces to the standard FJ model.

To facilitate analysis, the memory-augmented system can be equivalently expressed in a first-order state-space form. Define

$$\boldsymbol{u}^{(t)} = \boldsymbol{z}^{(t-1)}, \qquad \boldsymbol{y}^{(t)} = \begin{bmatrix} \boldsymbol{z}^{(t)} \\ \boldsymbol{u}^{(t)} \end{bmatrix}.$$

Then the evolution of $\boldsymbol{y}^{(t)}$ is governed by

$$\boldsymbol{y}^{(t+1)} = \boldsymbol{G}\boldsymbol{y}^{(t)} + \boldsymbol{c},$$

where

$$\boldsymbol{G} = \begin{bmatrix} \eta(\boldsymbol{I}+\boldsymbol{D})^{-1}\boldsymbol{A} & (1-\eta)(\boldsymbol{I}+\boldsymbol{D})^{-1}\boldsymbol{A} \\ \boldsymbol{I} & \boldsymbol{0} \end{bmatrix}, \qquad \boldsymbol{c} = \begin{bmatrix} (\boldsymbol{I}+\boldsymbol{D})^{-1}\boldsymbol{s} \\ \boldsymbol{0} \end{bmatrix}.$$

The convergence of the proposed second-order iteration is therefore characterized by the spectral radius $\rho(\boldsymbol{G})$, which determines the worst-case convergence rate and forms the basis for the subsequent theoretical analysis.

THEOREM 5.1. *Let A be an averaging matrix with a real spectrum. Suppose that $x_i(1) = y_i$, $i \in \{1, 2, \dots, n\}$, and $\sum_{i=1}^n x_i(0) = \sum_{i=1}^n y_i$. Then the opinion vector $\boldsymbol{z}(t)$ of the second-order iteration converges, and faster than the original FJ iteration if $0 < \gamma < \sigma_2^2$.*

PROOF OF THEOREM 1 (ITEM 4). Let A be an averaging matrix with real spectrum, and consider the augmented iteration matrix

$$B = \begin{bmatrix} (g+1)A & -gI \\ I & 0 \end{bmatrix}, \qquad -1 < g < 1.$$

Denote by $\rho_2(B)$ the convergence factor of the augmented system (4), and by $\sigma_2 \in [0,1)$ the second largest singular value of A, which is also the convergence factor of the original system (2). Item 4 asserts that the augmented system converges strictly faster than the original one whenever

$$0 < g < \sigma_2^2, \qquad \text{i.e.,} \qquad \rho_2(B) < \sigma_2.$$

We first consider the case $-1 < g \le g^*$. By Lemma 4, the convergence factor of the augmented system is given by

$$\rho_2(B) = \frac{1}{2}\Big(\sigma_2(g+1) + \sqrt{\sigma_2^2(g+1)^2 - 4g}\Big),$$

and in particular $\rho_2(B) = \sigma_2$ when $g = 0$. Moreover, Proposition 3 shows that $\rho_2(B)$ is strictly decreasing on the interval $(-1, g^*]$. Consequently, for any

$$0 < g \le \min\{g^*, \sigma_2^2\},$$

we have

$$\rho_2(B) < \rho_2(B)\big|_{g=0} = \sigma_2.$$

Next, consider the case $g^* < g < 1$. By Lemma 5, the convergence factor simplifies to

$$\rho_2(B) = \sqrt{g}.$$

Therefore, for any g satisfying

$$g^* < g < \sigma_2^2,$$

it follows immediately that

$$\rho_2(B) = \sqrt{g} < \sigma_2.$$

Combining the two cases above, we conclude that for all

$$0 < g < \sigma_2^2,$$

the inequality

$$\rho_2(B) < \sigma_2$$

holds. Hence, the augmented system (4) converges strictly faster than the original system (2), which completes the proof of Item 4. □

PROPOSITION 5.2. *Suppose that A is an matrix with a real spectrum. Let*

$$\boldsymbol{B} = \begin{bmatrix} (\gamma+1)A & -\gamma\boldsymbol{I} \\ \boldsymbol{I} & 0 \end{bmatrix}$$

where $-1 < \gamma < 1$. Then the minimum of $\rho_2(B)$ is unique. The iteration converge fastest when

$$\gamma = \frac{1 - \sqrt{1-\sigma_2^2}}{1 + \sqrt{1-\sigma_2^2}}. \tag{10}$$

Proof of Proposition 3. From Lemmas 4 and 5, it is clear that $\rho_2(B)$ is a continuous function of γ on the interval $(-1, 1)$. When $-1 < \gamma \leq \gamma^*$, by Lemma 4 and the fact that $\partial\rho_2/\partial\gamma < 0$, $\rho_2(B)$ is a decreasing function of γ. When $\gamma^* < \gamma < 1$, $\rho_2(B)$ is an increasing function of γ by Lemma 5. Since $\rho_2(B)$ is continuous at γ^*, γ^* is the unique point in the interval $(-1, 1)$ which minimizes $\rho_2(B)$. Then the optimal value of γ at this minimum is

$$\gamma^* = \frac{1 - \sqrt{1-\sigma_2^2}}{1 + \sqrt{1-\sigma_2^2}},$$

and the minimum of $\rho_2(B)$ is

$$\rho_2^* = \frac{\sigma_2}{1 + \sqrt{1-\sigma_2^2}}. \tag{11}$$

□

Algorithm 3: SecondOrderIter(G, γ, ϵ, T_{max})

Input : A graph $G = (V, E)$ with adjacency matrix A, degree vector d, internal opinions s; acceleration parameter $\gamma \in \mathbb{R}$; tolerance ϵ and maximum iterations T_{max}.
Output: Estimated opinion vector z.

Precompute:
$\widehat{D}^{-1} \leftarrow \mathrm{diag}\left(\frac{1}{1+d_i}\right)_{i\in V}$; $P \leftarrow \widehat{D}^{-1}A$; $b \leftarrow \widehat{D}^{-1}s$
Initialize:
$z^{(0)} \leftarrow s$; $k \leftarrow 0$; $r \leftarrow +\infty$
while $k < T_{max}$ ***and*** $r > \epsilon$ **do**
  $u \leftarrow P\,z^{(k)}$ // u is a work vector
  $z^{(k+1)} \leftarrow \gamma\, u + (1-\gamma)\, z^{(k-1)} + b$
  $r \leftarrow \| z^{(k+1)} - z^{(k)} \|_2$
  if $r \leq \epsilon$ **then**
    return $z^{(k+1)}$
  $z^{(k-1)} \leftarrow z^{(k)}$; $z^{(k)} \leftarrow z^{(k+1)}$
  $k \leftarrow k + 1$
return $z^{(k)}$

6 Experiments

In this section, we assess the efficiency and accuracy of our estimation algorithm LatestIter and MemoIter. To this end, we implement this algorithm on various real networks and compare the running time and accuracy of our algorithm with those corresponding to the previous algorithm, called LocalIter. The estimation of relevant quantities perform product of related matrices, and calculate corresponding ℓ_2 norms.

Environment. Our code for the estimation algorithm LatestIter and MemoIter were written in *Julia v1.5.1*. The *Julia language* implementation for which is open and accessible on the website[1].

Datasets and Equipment All the real-world networks we consider are publicly available in the Koblenz Network Collection and Network Repository. Our experiments are conducted on a diverse range of both undirected and directed networks, including social networks. For these datasets, the number n of nodes ranges from about 16 thousand to 33 million, and the number m of directed edges ranges from about 25 thousand to 301 million. The details of these datasets are presented in Table 1. All experiments are conducted using the Julia programming language. We conduct all experiments in a computational environment featuring a 4.2 GHz Intel i7-7700 CPU with 64GB of primary memory.

Algorithm For evaluating the performance of our algorithms estimating the opinion vector of the FJ model, we compare our three proposed algorithms LatestIter,MemoIter , and SecondIter with two state-of-the-art algorithms, namely Solver and BLI Sor. While one of the two state-of-the-art algorithms, Solver, is confined to undirected graphs due to the limitations of the Laplacian solver. Our proposed algorithm are applicable to both undirected and directed graphs.

Internal opinion distributions. In our experiments, the internal opinions are generated according to three different distributions: uniform distribution, exponential distribution, and power-law distribution. For the uniform distribution, the opinion s_i of node i is distributed uniformly in the range of $[0, 1]$. For the exponential distribution, we choose the probability density $e^{x_{\min}}e^{-x}$ to generate n' positive numbers x with minimum value $x_{\min} > 0$. Then, dividing each x by the maximum observed value, we normalize these n' numbers to the range $[0, 1]$ as the internal opinions of nodes. Note that there is always a node with internal opinion 1 due to the normalization operation. Similarly, for the power-law distribution, we use the probability density $(\alpha-1)x_{\min}^{\alpha-1}x^{-\alpha}$ with $\alpha = 2.5$ to generate n' positive numbers, and normalize them to interval $[0, 1]$ as the internal opinions.

Efficiency. Table 2 reports the running time of six methods, namely Solver, LI, LISOR, LatestIter, MemoIter, and SecondIter, on several real-world networks of increasing scale. We observe that Solver, which computes the exact solution by directly solving the linear system, is consistently the most time-consuming method and becomes increasingly expensive as the network size grows. In contrast, all iterative methods exhibit significantly better scalability.

Among the iterative solvers, LI and LISOR show stable and predictable performance across all datasets, with LISOR being slightly faster than LI in most cases. The Gauss–Seidel based method LatestIter and its over-relaxed variant MemoIter further reduce the running time on medium-scale networks such as Enron and DBLP, while maintaining low iteration counts. The second-order accelerated method SecondIter is the fastest on several large graphs, including

[1]URL omitted here.

Table 1: Statistics of the evaluated datasets.

Dataset	n	m	d_{max}
BlogCatalog	2,426	16,630	273
DBLP	317,080	1,049,866	343
YouTubeSnap	1,134,890	2,987,624	28,754
Flixster	2,523,386	7,918,801	1,474
out.youtube-u-growth	3,072,441	117,185,083	33,313

YouTubeSnap and out.youtube, where it achieves the shortest running time among all tested methods.

Overall, Table 2 demonstrates that the proposed iterative methods provide substantial efficiency gains over the exact solver, especially on large-scale graphs with millions of nodes, where SOLVER becomes prohibitively slow, while the iterative methods remain practical.

Accuracy. Besides efficiency, we also evaluate the accuracy of the iterative methods by comparing their outputs with the exact solution produced by SOLVER. For each method and each dataset, we report the absolute error on a representative entry of the solution vector, as shown in the right part of Table 2.

The results indicate that LI and LISOR consistently achieve errors on the order of 10^{-3} or below across all datasets, demonstrating reliable approximation quality. LATESTITER and MEMOITER generally attain even smaller errors, often in the range of 10^{-5} to 10^{-4}, especially on moderately sized networks such as Enron and DBLP. Although SECONDITER occasionally exhibits slightly larger errors on some large graphs (e.g., DBLP and out.youtube), its error remains within acceptable bounds for practical applications.

Overall, the empirical results confirm that all iterative methods achieve high accuracy in practice, with errors that are small

7 Conclusion

In this paper, we addressed the problem of computing single-node equilibrium opinions in the Friedkin–Johnsen model. We introduced the absorbing random-walk interpretation and developed local algorithms that avoid full-matrix computation. Our first contribution is a residual-based framework with synchronous and asynchronous iterations, together with a residue invariant that precisely characterizes the error and enables practical stopping rules and convergence bounds.

To further enhance scalability, we proposed a push–sampling algorithm (AIMS) that couples asynchronous local pushes with importance sampling over the residual distribution. The method preserves the invariant, uses random walks only to estimate the unprocessed mass, and yields an unbiased estimator with concentration guarantees. The τ-controlled sampling budget provides a clear trade-off between push work and variance, making the approach effective on both directed and undirected graphs.

Extensive experiments on real networks with up to tens of millions of nodes demonstrate that our methods are accurate and efficient, outperforming classical solvers and direct Monte Carlo baselines, and extending naturally to average-opinion estimation.

Currently, our framework does not adaptively schedule the mix of push and sampling across nodes. In future work, we plan to develop variance-aware scheduling, extend to time-varying and weighted networks, and integrate our methods with preconditioned linear solvers to support batch queries.

Table 2: Running time (seconds) and error of different iterative methods on real-world networks.

Dataset	Running time (s)						Error				
	Solver	LI	LISOR	LatestIter	MemoIter	SecondIter	LI	LISOR	LatestIter	MemoIter	SecondIter
BlogCatalog	5.08	0.54	0.20	0.08	0.02	0.47	1.29×10^{-3}	5.62×10^{-4}	3.44×10^{-2}	7.09×10^{-2}	1.38×10^{-3}
Enron	0.37	0.02	0.01	0.03	0.01	0.01	3.13×10^{-1}	3.12×10^{-1}	5.11×10^{-3}	1.78×10^{-3}	8.74×10^{-2}
DBLP	2.37	0.66	0.16	0.39	0.09	0.16	1.28×10^{-3}	4.79×10^{-4}	3.19×10^{-5}	8.76×10^{-5}	2.24×10^{-2}
YouTubeSnap	7.19	2.23	0.56	0.65	0.35	0.21	1.30×10^{-3}	2.74×10^{-4}	1.78×10^{-3}	8.43×10^{-4}	1.44×10^{-2}
Flixster	17.41	7.64	1.88	1.72	0.94	3.19	1.35×10^{-3}	2.97×10^{-4}	3.08×10^{-3}	3.35×10^{-4}	3.89×10^{-3}
out.youtube	26.64	12.02	2.33	3.75	1.39	0.84	1.09×10^{-3}	5.35×10^{-5}	3.46×10^{-4}	5.13×10^{-4}	8.37×10^{-3}

A Appendix: Pseudocode

In Appendix A, we present eliminated pseudocode in the main text.

B Appendix: Proof

In Appendix B, we provide the proofs of the theorems and lemmas in the main text.

B.1 Useful Tools

B.1.1 (matrix convergence).

Lemma B.1. *Let $B = (b_{ij})_{n\times n}$. Then $B^k \to 0$ as $k \to \infty$ if and only if $\rho(B) < 1$.*

Theorem B.2 (Error Bound). *For any accuracy parameter $\epsilon > 0$, the estimator $\boldsymbol{z}^{(t)}$ returned by the algorithm satisfies*

$$\|\boldsymbol{z}^* - \boldsymbol{z}^{(t)}\|_2 \le \epsilon. \tag{12}$$

Proof. From Lemma 1, we have $\boldsymbol{z}^* - \boldsymbol{z}^{(t)} = (\boldsymbol{I}+\boldsymbol{L})^{-1}\boldsymbol{r}^{(t)}$. Taking the ℓ_2-norm on both sides gives

$$\begin{aligned}\|\boldsymbol{z}^* - \boldsymbol{z}^{(t)}\|_2 &= \|(\boldsymbol{I}+\boldsymbol{L})^{-1}\boldsymbol{r}^{(t)}\|_2 \\ &\le \|(\boldsymbol{I}+\boldsymbol{L})^{-1}\|_2\, \|\boldsymbol{r}^{(t)}\|_2. \end{aligned} \tag{13}$$

By the termination criterion of the algorithm, the residual norm satisfies $\|(\boldsymbol{I}+\boldsymbol{L})^{-1}\|_2\, \|\boldsymbol{r}^{(t)}\|_2 \le \epsilon$, which completes the proof. $\square$

Lemma B.3.

$$\|\boldsymbol{z}^* - \boldsymbol{z}^{(t+1)}\|_v \le \left\|(\boldsymbol{I} - (\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})^{-1}\boldsymbol{A}_{up})^{-1}\right\|_v\, \|\boldsymbol{z}^{(t+1)} - \boldsymbol{z}^{(t)}\|_v, \tag{14}$$

$$\boldsymbol{z} - \boldsymbol{z}^{(t)} = (\boldsymbol{I}+\boldsymbol{L})^{-1}\boldsymbol{r}^{(t)}$$

$$\boldsymbol{z}^{(t+1)} - \boldsymbol{z}^{(t)} = (\boldsymbol{I}+\boldsymbol{D})^{-1}\boldsymbol{r}^{(t)}$$

$$\boldsymbol{z} - \boldsymbol{z}^{(t)} = (\boldsymbol{I}+\boldsymbol{L})^{-1}(\boldsymbol{I}+\boldsymbol{D})(\boldsymbol{z}^{(t+1)} - \boldsymbol{z}^{(t)})$$

Proof. The latest-opinion update rule can be written as $\boldsymbol{z}^{(t+1)} = (\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})^{-1}\boldsymbol{A}_{up}\boldsymbol{z}^{(t)} + (\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})^{-1}\boldsymbol{s}$. Define the successive difference $\Delta\boldsymbol{z}^{(t+1)} = \boldsymbol{z}^{(t+1)} - \boldsymbol{z}^{(t)}$. Then it follows directly that $\Delta\boldsymbol{z}^{(t+1)} = (\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})^{-1}\boldsymbol{A}_{up}\,\Delta\boldsymbol{z}^{(t)}$.

Since the iteration converges to the unique fixed point $\boldsymbol{z}^*$, we have $\boldsymbol{z}^* - \boldsymbol{z}^{(t+1)} = \sum_{k=t+1}^{\infty}\Delta\boldsymbol{z}^{(k+1)}$. Substituting the recursive relation of $\Delta\boldsymbol{z}^{(t)}$ yields

$$\begin{aligned}\boldsymbol{z}^* - \boldsymbol{z}^{(t+1)} &= \sum_{l=0}^{\infty}\left[(\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})^{-1}\boldsymbol{A}_{up}\right]^l \Delta\boldsymbol{z}^{(t+1)} \\ &= \left(\boldsymbol{I} - (\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})^{-1}\boldsymbol{A}\right)^{-1}\Delta\boldsymbol{z}^{(t+1)}, \end{aligned} \tag{15}$$

where the inverse exists due to the convergence of the Gauss–Seidel iteration.

Taking the ℓ_1-norm on both sides leads to

$$\|\boldsymbol{z}^* - \boldsymbol{z}^{(t+1)}\|_1 \le \left\|(\boldsymbol{I} - (\boldsymbol{I}+\boldsymbol{D})^{-1}\boldsymbol{A})^{-1}\right\|_1 \|\boldsymbol{z}^{(t+1)} - \boldsymbol{z}^{(t)}\|_1, \tag{16}$$

which establishes the claimed relation between the true estimation error and the difference of two consecutive iterates. $\square$

Lemma B.4 (Equivalent A Posteriori Error Representation). *Let $\boldsymbol{z}^{(t)}$ be the sequence generated by LatestIter and let $\boldsymbol{B} = (\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})^{-1}\boldsymbol{A}_{up}$ denote the corresponding iteration matrix. Then, for any vector norm $\|\cdot\|_v$ compatible with matrix multiplication, the following relation holds:*

$$\|\boldsymbol{z}^* - \boldsymbol{z}^{(t+1)}\|_v \le \|(\boldsymbol{I}-\boldsymbol{B})^{-1}\|_v\, \|\boldsymbol{z}^{(t+1)} - \boldsymbol{z}^{(t)}\|_v. \tag{17}$$

Proof. The LatestIter update can be written as

$$\boldsymbol{z}^{(t+1)} = \boldsymbol{B}\boldsymbol{z}^{(t)} + (\boldsymbol{I}+\boldsymbol{D}-\boldsymbol{A}_{low})^{-1}\boldsymbol{s}.$$

Define the successive difference $\Delta\boldsymbol{z}^{(t+1)} = \boldsymbol{z}^{(t+1)} - \boldsymbol{z}^{(t)}$. Then

$$\Delta\boldsymbol{z}^{(t+1)} = \boldsymbol{B}\,\Delta\boldsymbol{z}^{(t)}.$$

Since the iteration converges to the unique fixed point $\boldsymbol{z}^*$, we have

$$\boldsymbol{z}^* - \boldsymbol{z}^{(t+1)} = \sum_{k=t+1}^{\infty}\Delta\boldsymbol{z}^{(k+1)}.$$

Substituting the recursive relation of $\Delta\boldsymbol{z}^{(t)}$ yields

$$\boldsymbol{z}^* - \boldsymbol{z}^{(t+1)} = \sum_{l=0}^{\infty}\boldsymbol{B}^{\,l}\,\Delta\boldsymbol{z}^{(t+1)} = (\boldsymbol{I}-\boldsymbol{B})^{-1}\Delta\boldsymbol{z}^{(t+1)},$$

where the inverse exists since the Gauss–Seidel iteration converges and $\rho(\boldsymbol{B}) < 1$.

Taking the $\|\cdot\|_v$ norm on both sides and using submultiplicativity completes the proof. $\square$